A Spatially Adaptive Finite State Projection for the Reaction-Diffusion Master Equation

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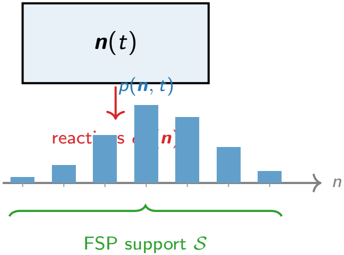
The Chemical Master Equation (CME)

A **single well-mixed compartment** holding molecules of S species. State: molecule count vector $\boldsymbol{n} \in \mathbb{Z}^S_{\geq 0}$.

Each reaction r fires with rate $\alpha_r(\boldsymbol{n})$ and changes the state by stoichiometry $\boldsymbol{\nu}_r$. The probability distribution $p(\boldsymbol{n}, t)$ evolves as

$$\frac{dp}{dt} = \underbrace{Ap}_{\text{CME}},$$

where $A_{ij} = \alpha_r(\boldsymbol{n}_j)$ if $\boldsymbol{n}_i = \boldsymbol{n}_j + \boldsymbol{\nu}_r$, and the diagonal ensures column sums are zero.



Finite State Projection (FSP)

Truncate to a finite support $\mathcal{S}$; expand it adaptively as probability mass spreads. State space $|\mathcal{S}|$ grows only as needed — not exponentially in time.

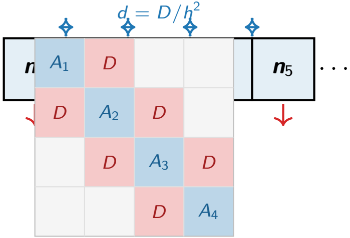
The RDME: K coupled CMEs

Divide the spatial domain into K voxels of width h. Each voxel k has its own count vector $\boldsymbol{n}_k$ and **its own CME**.

Voxels are coupled by **diffusion**: one molecule hops $k \leftrightarrow k\pm1$ at rate $d = D/h^2$ per molecule.

The joint distribution $p(\boldsymbol{n}_1, \ldots, \boldsymbol{n}_K, t)$ obeys

$$\frac{dp}{dt} = \underbrace{(A_{\mathrm{rxn}} + A_{\mathrm{diff}})}_{A_{\mathrm{RDME}}}\, p.$$



Exponential cost

Full state space: $n_{\max}^{SK}$.
Even $K{=}8$, $S{=}1$, $n_{\max}{=}50$: intractable.

A_{RDME} is **block-tridiagonal**: A_k on diagonal (CME), $D_{k,k+1}$ off-diagonal (diffusion shifts one

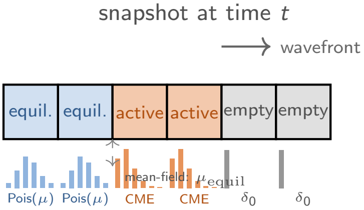
Key observation: molecules don't teleport

Starting from a **localised initial condition** (all molecules in one region), probability mass spreads *diffusively*.

At any time t, the support of p is concentrated near the **diffusion wavefront**: voxels far from the origin are essentially empty.

Three regimes at any instant:

- **Empty**: $p(\boldsymbol{n}_k) \approx \delta_{\boldsymbol{n}_k,\mathbf{0}}$. No tracking needed.
- **Active**: wavefront voxels. Nontrivial distribution — track with the full CME.
- **Equilibrated**: behind the wavefront, distribution has converged to a known form. Store as a single parameter.



Mean-field coupling: exact for linear kinetics

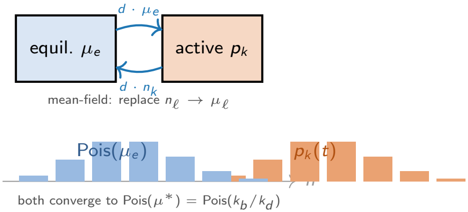
Exact RDME: gain rate into voxel k from neighbor ℓ is $d \cdot n_\ell$, which depends on the *random* count n_ℓ.

Mean-field approximation: replace the random n_ℓ by its mean $\mu_\ell = \mathbb{E}[n_\ell]$, decoupling the voxel CMEs:

$$\frac{dp_k}{dt} = Q_k(\mu_{\rm nbr})\, p_k,$$

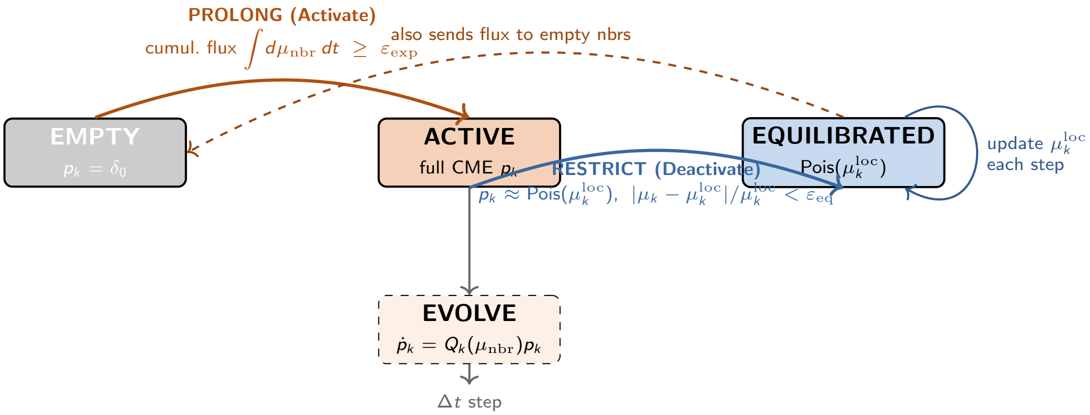
where the effective generator has

$$k_b^{\rm eff} = k_b + d \sum_{\ell \sim k} \mu_\ell, \qquad k_d^{\rm eff} = k_d + d\,|\text{nbrs}|.$$



When is mean-field exact?

For **linear (birth-death) kinetics**, the SS is a product of independent Poissons: $p^* = \prod_k \text{Pois}(\mu^*)$. Detailed balance holds under symmetric diffusion, so



Why is μ_k^{loc} updated each step? The equilibrated voxel's local SS depends on its neighbours: $\mu_k^{\mathrm{loc}}(t) = (k_b + d\sum_\ell \mu_\ell)/(k_d + d|\mathrm{nbrs}|)$. As active neighbours equilibrate, $\mu_k^{\mathrm{loc}} \rightarrow k_b/k_d$ (global SS).

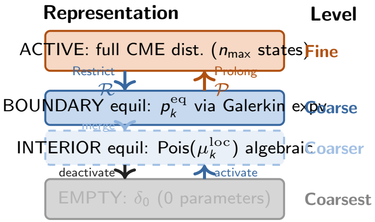
Connection to multigrid: an adaptive spatial V-cycle

Classical geometric multigrid (for linear PDEs):

1. *Pre-smooth*: relax on fine grid.
2. *Restrict* $\mathcal{R}$: fine $\to$ coarse.
3. *Coarse solve*: exact solve on coarse grid.
4. *Prolong* $\mathcal{P}$: coarse $\to$ fine.
5. *Post-smooth*: relax on fine grid.

Spatial adaptive FSP (this work):

1. *Evolve*: CME step on active (fine) voxels.
2. *Restrict* $\mathcal{R}$: active $\to$ equil; store p_k as p_k^{eq}.
3. *Coarse update* ($A_c = \mathcal{R}A\mathcal{P}$, Galerkin): boundary equil $\to$ expv; interior equil $\to$ algebraic.
4. *Prolong* $\mathcal{P}$: flux threshold $\to$ activate.



Four levels: active / boundary-equil / interior-equil / empty. Galerkin coarse operator $A_c = \mathcal{R}A\mathcal{P}$ used for boundary equil; algebraic mean for interior (no matrix exp).

What makes step 3 Galerkin?

A_c is obtained by marginalising A_{RDME} over independent neighbours (mean field). For birth–death

1. **Compute effective means.** Active: $\mu_k = \langle p_k \rangle$. Equil: $\mu_k = \langle p_k^{\mathrm{eq}} \rangle$ (from stored dist.).
2. **Evolve active voxels** (full CME, Krylov m). $p_k \leftarrow e^{Q_k \Delta t} p_k, \;\; Q_k = Q(k_b + d \sum_\ell \mu_\ell, \; k_d + d|\mathrm{nb}|)$.
3. **Update equilibrated voxels** (Galerkin $A_c = \mathcal{R}A\mathcal{P}$). *Boundary* ($\exists$ active nb): $p_k^{\mathrm{eq}} \leftarrow e^{Q_k^{\mathrm{eq}} \Delta t} p_k^{\mathrm{eq}}$, small Krylov $m_{\mathrm{eq}} \ll m$. *Interior* (all nb equil/empty): $p_k^{\mathrm{eq}} \leftarrow \mathrm{Pois}(\mu_k^{\mathrm{loc}})$, algebraic.
4. **Accumulate flux** into empty voxels from active+equil. $\Phi_k \mathrel{+}= d\,\mu_\ell\,\Delta t$.
5. **Prolong (activate).** $\Phi_k \geq \varepsilon_{\exp} \Rightarrow p_k = \delta_0$.
6. **Restrict (deactivate).** $|\mu_k - \mu_k^{\mathrm{loc}}| / \mu_k^{\mathrm{loc}} < \varepsilon_{\mathrm{eq}}$ **and** $d_{\mathrm{TV}}(p_k, \mathrm{Pois}(\mu_k^{\mathrm{loc}})) < \varepsilon_{\mathrm{TV}}$: set $p_k^{\mathrm{eq}} \leftarrow p_k$ (no information lost).

Cost per step (three tiers)

Tier	Count	Work
Active	K_{act}	$\mathrm{expv}(n_{\max}, m)$
Boundary equil	$O(\sqrt{K})$	$\mathrm{expv}(n_{\mathrm{eq}}, m_{\mathrm{eq}})$
Interior equil	$O(K_{\mathrm{eq}})$	$O(n_{\mathrm{eq}})$ algebraic

$K_{\mathrm{act}} \ll K$ (wavefront ring only). $m_{\mathrm{eq}} \ll m$ (near-Poisson $\Rightarrow$ fast Krylov). Interior equil: no matrix exp at all.

State-space comparison

Method	States tracked
Full FSP	$n_{\max}^{SK}$
Spatial adaptive FSP	$K_{\mathrm{act}} \cdot n_{\max}^{S}$

Galerkin coarse operator: boundary vs. interior equilibrated voxels

The Galerkin coarse operator for a single voxel k:

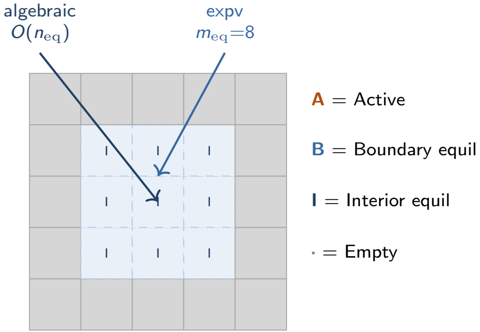
Marginalise A_{RDME} over all other voxels assuming independence (mean-field $n_\ell \to \mu_\ell$):

$$A_c^{(k)} = \underbrace{Q\big(k_b + d \textstyle\sum_\ell \mu_\ell,\ k_d + d|\mathrm{nb}|\big)}_{\text{single-voxel birth-death CME with eff. rates}} .$$

This is *exactly* what `build_coarse_system` derives for one voxel of the coarsened RDME.

Two update rules depending on neighbourhood:

Boundary	**Interior**
≥ 1 active nb	all nb equil/empty
$p_k^{\mathrm{eq}} \leftarrow e^{A_c^{(k)} \Delta t} p_k^{\mathrm{eq}}$	$p_k^{\mathrm{eq}} \leftarrow \mathrm{Pois}(\mu_k^{\mathrm{loc}})$
Krylov $m_{\mathrm{eq}}{=}8$	algebraic, $O(n_{\mathrm{eq}})$
Rates change fast	Rates change slowly



At any instant, **boundary equil** $= O(\sqrt{K})$ voxels (wavefront circumference). Interior equil $= O(K)$ but costs only $O(n_{\mathrm{eq}})$ each — no matrix exponential.

2D birth-death: wavefront propagation and collapse

Model. $\emptyset \xrightarrow{k_b} A \xrightarrow{k_d} \emptyset$ in each voxel with diffusion D. Steady state: each voxel independent $\mathrm{Pois}(k_b/k_d)$.

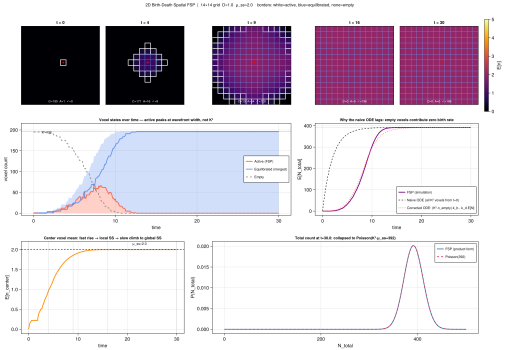
IC. All voxels empty; center voxel activated at $t = 0$. Birth reactions seed the wavefront.

Parameters: $K{=}14$, $k_b{=}1$, $k_d{=}0.5$, $D{=}1$, $\mu^*{=}2$.

Voxel state counts over time:

- Active count peaks at $\sim$66 (one wavefront ring), **not** $K^2 = 196$.
- Equilibrated count grows monotonically behind the front.
- Active $\to$ 0: the distribution is *fully represented by coarse parameters*.

ODE comparison. Naive $K^2 k_b$ ODE *over*estimates: empty voxels do not birth. Corrected ODE



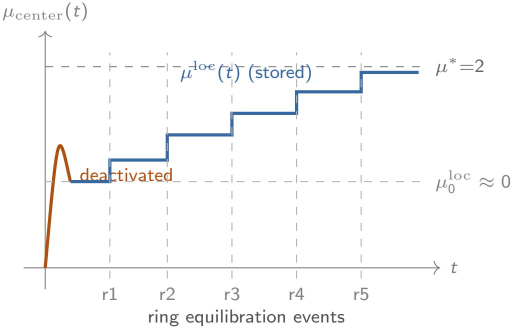
Center voxel: two-phase equilibration

The center voxel is activated at $t = 0$ and deactivated **early** once its distribution reaches its *local* SS.

Local SS (4 empty neighbours):

$$\mu^{\mathrm{loc}}_{\mathrm{center}} = \frac{k_b}{k_d + 4d} = \frac{1}{0.5 + 4} \approx 0.22.$$

The center's stored mean then **evolves upward** as ring-1, ring-2, . . . equilibrate behind the wavefront.



Two phases of the center mean:

1. **Fast:** $\mu \nearrow 0.22$ in $\sim 1/k_d^{\mathrm{eff}} \approx 0.2$ time units. Deactivated.
2. **Slow:** $\mu^{\mathrm{loc}} \nearrow 2.0$ over $\tau_{\mathrm{diff}} \approx (K/2)^2/(2D) \approx 24$ time units as successive rings equilibrate.

This **staircase** in μ^{loc} traces the equilibration wave

Collapse to a single well-mixed CME

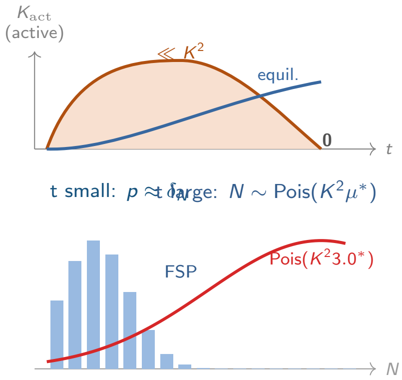
Once all voxels are equilibrated:

- All $\mu_k^{\mathrm{loc}} \to \mu^* = k_b/k_d = 2$.
- No active voxels remain; FSP support $= \emptyset$.
- Total count $N = \sum_k n_k$ is independent of spatial arrangement.

The **collapsed CME** for N is:

$$\emptyset \xrightarrow{K^2 k_b} N \xrightarrow{k_d} \emptyset,$$

with SS distribution $N^* \sim \mathrm{Pois}(K^2\mu^*)$.



Product-form exactness

For birth-death, the voxels are independent at SS: $p^* = \prod_k \mathrm{Pois}(\mu^*)$. Total distribution is convolution of K^2 independent Poissons:

K^2

What we built:

- A spatially adaptive FSP for the RDME that maintains only a **sparse active set** — the current wavefront ring.
- Three-level representation: empty (δ_0), equilibrated ($\mathrm{Pois}(\mu^{\mathrm{loc}})$), active (full CME).
- Physics-driven prolong/restrict cycle: **an adaptive multigrid V-cycle in state space**.

Key properties:

- Computational work $\propto K_{\mathrm{act}} \cdot n_{\max}$, **not** $n_{\max}^{K}$.
- Mean-field coupling is exact at SS for linear kinetics.
- Equilibrated means evolve algebraically (local SS tracking).
- Total distribution collapses to $\mathrm{Pois}(K^2\mu^*)$.

Open directions:

1. **Nonlinear kinetics** (Schlögl, bistable networks).
   Mean-field is no longer exact. Need joint tracking at wavefront pairs (patch QSD).
2. **Joint wavefront tracking.**
   When K_{act} voxels are correlated, maintain a *joint* distribution rather than independent marginals.
3. **Adaptive re-activation.**
   Re-activate equilibrated voxels when perturbations arrive (e.g., front reflections, new injections).
4. **Higher dimensions and irregular grids.**
   The prolong/restrict logic generalises